HW 4 Solutions

Note: in these solutions I use indicator functions of sets, which are defined as follows: if A is a set, then the indicator function of A, denoted $\mathbf{1}_A$, is simply defined to equal 1 on A and 0 elsewhere. Thus,

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

1

a

By Bayes rule, the posterior distribution of θ given the observed value x of X is $f(\theta|x) \propto f(x|\theta)f(\theta)$. In our case, the prior distribution is $f(\theta) = 1/6$ for $\theta \in \{1, \ldots, 6\}$, and the distribution of X given θ is $f(x|\theta) = \theta \cdot \mathbf{1}_{[0,1/\theta]}(x)$. Thus

$$f(x|\theta)f(\theta) = \frac{1}{6} \cdot \theta \cdot \mathbf{1}_{[0,1/\theta]}(x)$$

for $\theta \in \{1, \ldots, 6\}$. To write this in a more useful form, note that $0 \leq x \leq 1/\theta$ is equivalent to $0 < \theta \leq 1/x$. Therefore,

$$\mathbf{1}_{[0,1/\theta]}(x) = \mathbf{1}_{(0,1/x]}(\theta),$$

and so

$$f(x|\theta)f(\theta) = \frac{\theta}{6}$$

if $\theta \in \{1, \ldots, 6\}$ and $\theta \leq 1/x$, and $= 0$ otherwise. Using the observed value $X = 0.3$, we get $0 < \theta \leq 10/3$, so $f(\theta|0.3)$ is nonzero for $\theta = 1, 2, 3$ and zero else. Note that since $(1 + 2 + 3)/6 = 1$, we don't have to normalize, and this is exactly the posterior distribution of θ. Clearly it is maximized when $\theta = 3$, so that is the MAP estimate of θ.

b

The distribution of three independent samples given the value for θ is

$$f(x_1, x_2, x_3|\theta) = f(x_1|\theta)f(x_2|\theta)f(x_3|\theta) = \theta^3 \cdot \mathbf{1}_{[0,1/\theta]^3}(x_1, x_2, x_3).$$

Multiplying by the prior, we get

$$f(\theta|x_1, x_2, x_3) \propto \frac{\theta^3}{6} \cdot \mathbf{1}_{[0,1/\theta]^3}(x_1, x_2, x_3)$$

for $\theta \in \{1, \ldots, 6\}$. In other words,

$$f(\theta|x_1, x_2, x_3) \propto \frac{\theta^3}{6}$$

for $\theta = 1, \ldots, 6$ and $0 < \theta \leq \min\{1/x_1, 1/x_2, 1/x_3\}$. In our case, the minimum is $1/X_2 = 1/0.4 = 2.5$. Thus, $\theta \in \{1, 2\}$. Again, the largest allowable value of θ obviously maximizes this posterior distribution, so the new MAP estimate of θ is 2.

2

Using the same notation as in problem 1, we have $f(x|\theta) = (2\pi 9)^{-1/2} \exp(-(x-\theta)^2/18)$, $f(\theta) = (2\pi 4)^{-1/2} \exp(-(\theta - 5)^2/8)$. Therefore

$$f(\theta|x) \propto \exp(-(x-\theta)^2/18 - (\theta-5)^2/8).$$

Since we observed $X = 1$,

$$f(\theta|1) \propto \exp(-(1-\theta)^2/18 - (\theta-5)^2/8) \propto \exp(-\frac{13}{72}(\theta - 49/13)^2).$$

Thus $\theta|X = 1 \sim N(49/13, 36/13)$. Therefore, since the normal pdf is greatest at its mean, the MAP and conditional expectation estimates of θ are both 49/13.

3

a

The formula is

$$[\bar{x} - t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}}].$$

For us $n = 21, \alpha = 0.1, \bar{x} = 72.3, s = \sqrt{5.6}, t_{0.05}(20) = 1.725$. Therefore our interval is [71.41,73.19].

b

This interval is not centered around the mean, so we have to calculate each side separately: [70.5,72.3] and [72.3,73]. The confidence of [70.5,72.3] is half the confidence of the symmetric version [70.5,74.1]. We have $72.3 - 70.5 = 1.8 = t_{\alpha/2}(20)\frac{s}{\sqrt{n}}$, so $t_{\alpha/2}(20) = 3.49$, which by the table in the back of the book corresponds to $\alpha/2 \approx 0$ (beyond the precision of the table), or about 100%. Therefore the confidence of [70.5,72.3] is about 50%. For the other, we get $t_{\alpha/2}(20) = 1.35$, which is just above the $t_{0.1}(20)$ mark, so the confidence is approximately 80%, which when halved becomes 40%. Therefore the total confidence is about 50%+40% = 90%.

c

The formula is

$$(-\infty, \bar{x} + t_\alpha(n-1)\frac{s}{\sqrt{n}}].$$

Plugging in $t_{0.1}(20) = 1.325$ gives $(-\infty, 72.98]$.

d

Using the same reasoning as above, with z_α replacing $t_\alpha(20)$ and 5 replacing $s^2 = 5.6$, we get
a. 72.3 ± 0.80,
b. $\sim 50\% + \sim (82.3/2)\% \approx 91\%$
c. $(-\infty, 72.3 + 1.13]$

4

a

$\bar{x} = 3.58$

b

$s = 0.5116422$ (sample standard deviation)

c

Since we are sampling from a normal distribution with unknown variance, we use the t-distribution. The formula is

$$(-\infty, \bar{x} + t_\alpha(n-1)\frac{s}{\sqrt{n}}].$$

Plugging in $t_{0.05}(9) = 1.833$ gives $(-\infty, 3.87657]$, so 3.87657 is an upper bound with confidence coefficient 0.95.

d

The formula is

$$[\bar{x} - t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2}(n-1)\frac{s}{\sqrt{n}}].$$

Plugging in $t_{0.1}(9) = 1.383$ gives 3.58 ± 0.223763.

5

$$\bar{X} - 1 < \mu < \bar{X} + 1 \iff -1 < \bar{X} - \mu < 1 \iff -\sqrt{n}/3 < \frac{\bar{X} - \mu}{3/\sqrt{n}} < \sqrt{n}/3.$$

Now $\bar{X} - \mu$ is $N(0, 9/n)$, with standard deviation $3/\sqrt{n}$, so $Z = (\bar{X} - \mu)/(3/\sqrt{n})$ is $N(0,1)$. Thus, we want n such that $\mathbb{P}(|Z| < \sqrt{n}/3) = 0.9$, i.e. we want $\sqrt{n}/3 = z_{(1-0.9)/2} = z_{0.05} = 1.645$, or $n = (3 * 1.645)^2 \approx 24.35$. Since we want a whole number, we round up to 25, so that we will be above 90% rather than below.

6

a

$\bar{d} = 0.07875$ (sample mean of the observed differences)

b

The formula is

$$[\bar{d} - t_\alpha(n-1)\frac{s}{\sqrt{n}}, \infty).$$

Plugging in $t_{0.05}(23) = 1.714$ gives $[-0.01043932, \infty)$, so the lower bound is -0.01043932.

c

For the 95% confidence level, the program doesn't appear to have been effective, as the lower bound is negative.